



Learnability and Persistence of Excitation of ILC

Gijo Sebastian, Jonathan Eden, and Ying Tan

The Department of Mechanical Engineering

The University of Melbourne

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Outline

- Learnability in ILC
- Learnability and PE Condition using Input Learnability
- Summary

Learnability in ILC



- In ILC, what does it mean for a task to be learnable?
- Arimoto and Naniwa [1] introduced learnability as the inherent ability to learn a desired motion

¹Arimoto, S. & Naniwa, T. (1992). "Learnability and Passivity for a Class of Mechanical Systems," Trans. SICE, 28(2), 237–246 (in Japanese)



Learnability in ILC

- For a given system and a given desired output trajectory $y_d(t)$, we distinguish two types of learnability:
 - ✓ Output learnability¹: Find a sequence of control input $\{u_i(t)\}_{i=1,2,\dots,t \in [0,T]}$ such that $\lim_{i \rightarrow \infty} y_i(t) = y_d(t), \forall t \in [0, T]$
 - ✓ Input learnability²: Suppose there exists a desired input $u_d(t)$ that generates $y_d(t)$. Find a sequence of control input $\{u_i(t)\}_{i=1,2,\dots,t \in [0,T]}$ such that $\lim_{i \rightarrow \infty} u_i(t) = u_d(t), \forall t \in [0, T]$

¹S. Arimoto, S. Kawamura, and F. Miyazaki, “Bettering operation of robots by learning,” *Journal of Robotic Systems*, vol. 1, no. 2, pp. 123–140, 1984

²B. J. Driessen and N. Sadegh, “Multi-Input Square Iterative Learning Control With Input Rate Limits and Bounds,” *IEEE Transactions on Systems, Man, and Cybernetics—Part B*, vol. 32, no. 4, pp. 545–550, 2002.



Learnability in ILC

➤ Are these two notions equivalent?

$$y_i(t) \rightarrow y_d(t) \quad \overset{??}{\longleftrightarrow} \quad u_i(t) \rightarrow u_d(t)$$

In general

$$u_i(t) \rightarrow u_d(t) \quad \longrightarrow \quad y_i(t) \rightarrow y_d(t)$$

$$y_i(t) \rightarrow y_d(t) \quad \not\longrightarrow \quad u_i(t) \rightarrow u_d(t)$$

Different control inputs may generate the same desired output¹⁻³

¹D. Meng, Y. Jia, J. Du, and F. Yu, “Two Alternative Approaches to Stochastic Discrete-Time Iterative Learning Control Systems,” Proceedings of the 50th IEEE CDC-ECC, 2011.

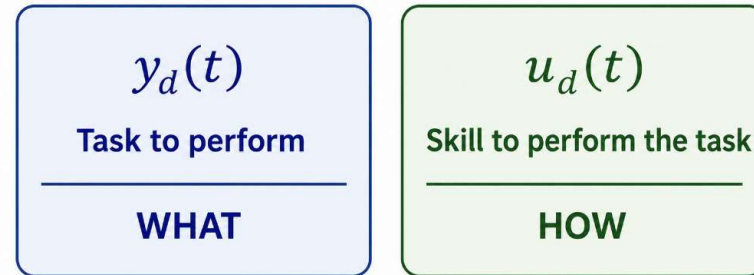
²G. Sebastian, Y. Tan, and D. Oetomo, “A Unified Analysis Tool in Iterative Learning Control: Composite Energy Function,” 12th Asian Control Conference (ASCC), pp. 1601–1606, 2019.

³D. Meng and K. L. Moore, “Convergence of Iterative Learning Control for SISO Nonrepetitive Systems Subject to Iteration-Dependent Uncertainties,” Automatica, vol. 79, pp. 167–177, 2017



Learnability in ILC

➤ Should we learn y_d or u_d ?



- ✓ Humans acquire and retain motor skills through practice¹⁻³ :“Practice makes perfect”
- ✓ Input learnability makes it possible to transfer the learned skill to a family of related tasks⁴

¹Krakauer, J. W., Hadjiosif, A. M., Xu, J., Wong, A. L., & Haith, A. M. (2019). “Motor Learning.” Comprehensive Physiology, 9(2), 613–663.

²Dayan, E., & Cohen, L. G., “Neuroplasticity Subservicing Motor Skill Learning.” Neuron, 72(3), 443–454, 2011.

³Shadmehr, R., & Brashers-Krug, T. , “Functional Stages in the Formation of Human Long-Term Motor Memory.” Journal of Neuroscience, 17(1), 409–419, 1997.

⁴J.-X. Xu, “Direct Learning of Control Efforts for Trajectories with Different Magnitude Scales,” Automatica, vol. 33, no. 12, pp. 2191–2195, 1997.



Learnability in ILC

- Because of trial-to-trial repeatability, we can adaptively learn the desired control input $u_d(t), t \in [0, T]$ pointwise across iterations
- This naturally leads to the concept of pointwise adaptation¹
- This establishes a natural connection with adaptive control and raises a fundamental question: Do we need persistent excitation (PE)² to learn $u_d(t), t \in [0, T]$

¹Y. Tan and J.-X. Xu, “A New Pointwise Adaptive Control Approach for Time-Varying Parameters with Known Periodicity,” Proceedings of the 2003 American Control Conference (ACC), vol. 2, pp. 1596–1601, 2003.

²K. S. Narendra and A. M. Annaswamy, “Persistent Excitation in Adaptive Systems,” International Journal of Control, vol. 45, no. 1, pp. 127–160, 1987.



Learnability and PE Condition using Input Learnability

➤ What is PE?

To learn an unknown quantity, the observed data must contain sufficient information about it

Consider a classic adaptive control law:

$$y(t) = \phi^T(t)\theta$$

To learn the unknown parameter $\theta \in R^m$, the information of $\phi(t)$ should be rich enough such that the following PE condition needs to be satisfied

$$\int_t^{t+T_0} \phi(\tau)\phi^T(\tau)d\tau \geq \alpha I_{m \times m} \forall t, \alpha > 0$$



Learnability and PE Condition using Input Learnability

- An example to illustrate this idea

$$y(t) = \theta_1 \phi_1(t) + \theta_2 \phi_2(t) \quad \text{If } \phi_2(t) = 2\phi_1(t), y(t) = (\theta_1 + 2\theta_2)\phi_1(t)$$

We cannot distinguish θ_1 and θ_2 individually

Not enough independent information $\rightarrow$ the parameters cannot be uniquely learned

- ✓ PE determines whether the unknown parameters are learnable; the strength/richness of excitation influences how fast they can be learned¹.
- ✓ More precisely, for a fixed learning law, a larger PE level α generally yields a faster guaranteed convergence rate

¹B. D. O. Anderson and C. R. Johnson Jr., "Exponential Convergence of Adaptive Identification and Control Algorithms," *Automatica*, vol. 18, no. 1, pp. 1–13, 1982.



Learnability and PE Condition using Input Learnability

- Through pointwise adaptation, we extend the classical PE concept for unknown parameters to the time-varying but iteration-invariant desired input $u_d(t)$

How to design ILC
algorithm to

guarantee learning of the unknown $u_d(t)$

accelerate the learning of $u_d(t)$

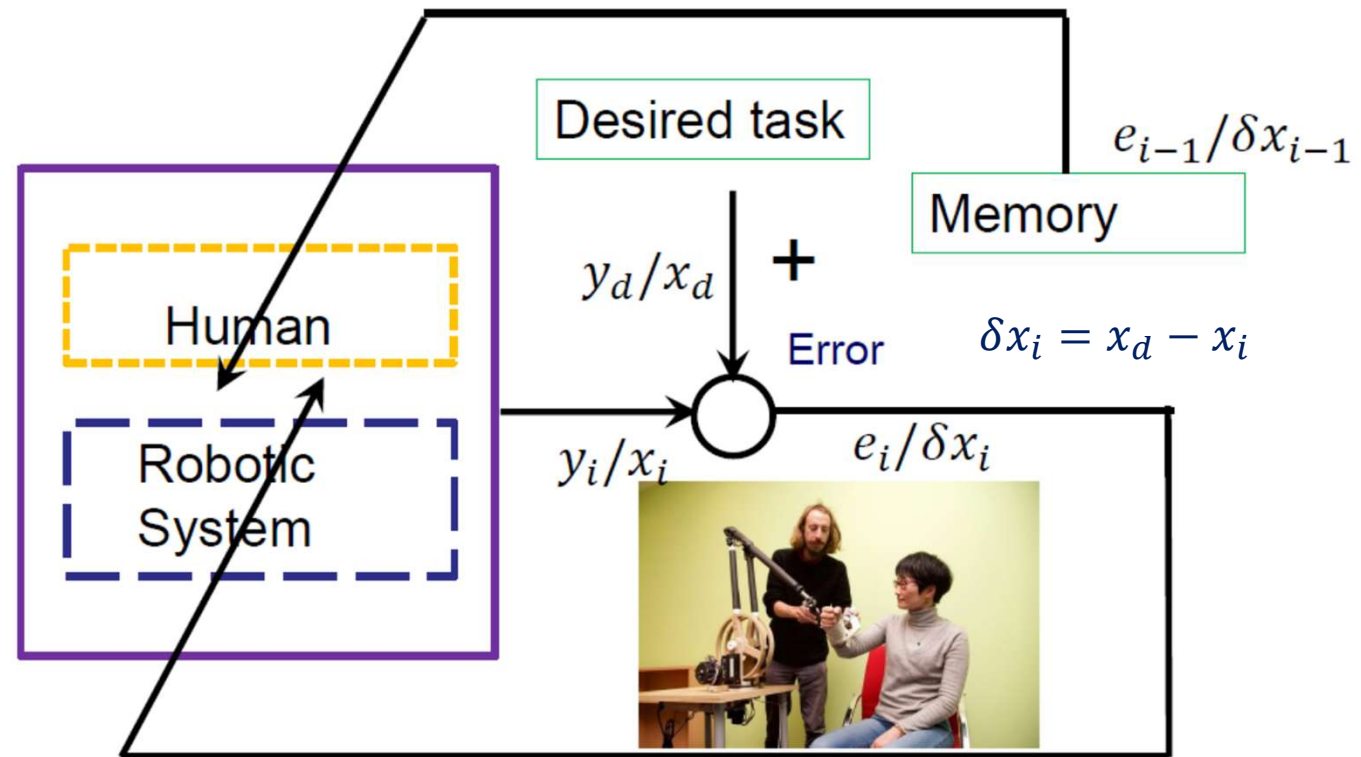
- How to generate PE condition? --By introducing the feedback

Learnability and PE Condition using Input Learnability

- A motivating example in a rehabilitation robotics setting

Human learns
to perform a
task iteratively

Robotic system
provides
feedback





Learnability and PE Condition using Input Learnability

- A motivating example in rehabilitation robotics setting:

$$\Sigma_i^{Overall}: \begin{cases} \dot{x}_i = \begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix} x_i + \begin{bmatrix} 0 \\ 1 \end{bmatrix} (u_i^H + u_i^R) \\ y_i = [1 \quad 0] x_i \end{cases} \quad x_i(0) = x_0, t \in [0, T]$$

- ✓ Assume that there exists a desired reference trajectory satisfying the following dynamics (matching condition)

$$\Sigma_d: \dot{x}_d = Ax_d + Bu_d, y_d = [1 \quad 0]x_d$$



Learnability and PE Condition using Input Learnability

➤ For a given y_d , the tracking error is $e_i(t) = y_d(t) - y_i(t)$

➤ The system has a relative degree of 2

➤ Assume that human learning follows a D-type ILC law:

$$u_i^H(t) = u_{i-1}^H(t) + \dot{e}_i(t)$$

➤ Assume that the robot uses a state-feedback controller to provide assistance

$$u_i^R(t) = K \Delta x_i(t) \quad \text{JE1} \quad \Delta x_i(t) = \begin{bmatrix} y_d - y_i \\ \dot{y}_d - \dot{y}_i \end{bmatrix}$$

Slide 13

JE1

If x_d is first then I believe this should be a positive sign (otherwise it won't be A-BK for Δx)

Jonathan Eden, 2026-08-21T01:37:10.944



Learnability and PE Condition using Input Learnability

- The closed-loop system becomes

$$\Delta \dot{x}_i(t) = (A - BK)\Delta x_i(t) + B(u_d(t) - u_i^H(t))$$

- We can show that such a feedback and feed-forward controller works for such a system by using the following composite energy function (CEF)

$$E_i(t) = e^{-\lambda t} \Delta x_i^T(t) \Delta x_i(t) + \int_0^t e^{-\lambda \tau} \left(\Delta u_i^H(\tau) \right)^2 d\tau$$

$$\Delta u_i^H(t) = u_i^H(t) - u_d(t) \text{ JE1}$$

Slide 14

JE1

To be consistent with earlier convention I believe this should be $u_d(t) - u_i^H(t)$

Jonathan Eden, 2026-08-21T01:38:16.589



Learnability and PE Condition using Input Learnability

- Under the resetting condition $\Delta x_i(0) = 0$, it has

Good term to cancel the dynamics

$$\begin{aligned}\Delta E_i(t) &= E_i(t) - E_{i-1}(t) \\ &= 2 \int_0^t e^{-\lambda\tau} \Delta x_i^T(\tau) \Delta \dot{x}_i(\tau) d\tau - \lambda \int_0^t e^{-\lambda\tau} \Delta x_i^T(\tau) \Delta x_i(\tau) d\tau \\ &\quad + \int_0^t e^{-\lambda\tau} \left(\left(\Delta u_i^H(\tau) \right)^2 - \left(\Delta u_{i-1}^H(\tau) \right)^2 \right) d\tau - e^{-\lambda t} \Delta x_{i-1}^T(t) \Delta x_{i-1}(t)\end{aligned}$$

Generate an adaptive update law to learn the unknown $u_d(t)$



Learnability and PE Condition using Input Learnability

- The design of the update law is similar to that in adaptive control

$$2\Delta x_i^T(\tau)\Delta\dot{x}(\tau) = 2\Delta x_i^T(\tau) \left((A - BK)\Delta x_i(t) + B(u_d(t) - u_{H,i}(t)) \right)$$

$$\begin{aligned} \left(\Delta u_i^H(\tau) \right)^2 - \left(\Delta u_{i-1}^H(\tau) \right)^2 &= (u_i^H - u_{i-1}^H)(u_i^H + u_{i-1}^H - 2u_d) \\ &= (u_i^H - u_{i-1}^H) \left(-(u_i^H - u_{i-1}^H) + 2(u_i^H - u_d) \right) \end{aligned}$$

If the updating law takes the following form:

$$u_i^H - u_{i-1}^H = \Delta x_i^T B = \dot{e}_i, \text{ JE1}$$

the influence of u_d in $\Delta\dot{x}$ in $\Delta E_i(t)$ will be cancelled. The design of the update law is similar to that in adaptive control

Slide 16

JE1

So that it is using existing knowledge (and so it is consistent with slide 13) this should be $\dot{e}_{i-1}$

Jonathan Eden, 2026-08-21T01:39:18.591



Learnability and PE Condition using Input Learnability

- We can show that the tracking error converges based on the following two observations
 - ✓ The first iteration, even if $(A - BK)$ is unstable, it is bounded within a finite time interval
 - ✓ $\Delta E_i(t)$ is non-positive by selecting a sufficiently large λ

$$\begin{aligned} & \Delta E_i(t) \\ & \leq (|A - BK| - \lambda) \int_0^t e^{-\lambda\tau} \Delta x_i^T(\tau) \Delta x(\tau) d\tau - e^{-\lambda t} \Delta x_{i-1}^T(t) \Delta x_{i-1}(t) \end{aligned}$$

JE1

Slide 17

JE1

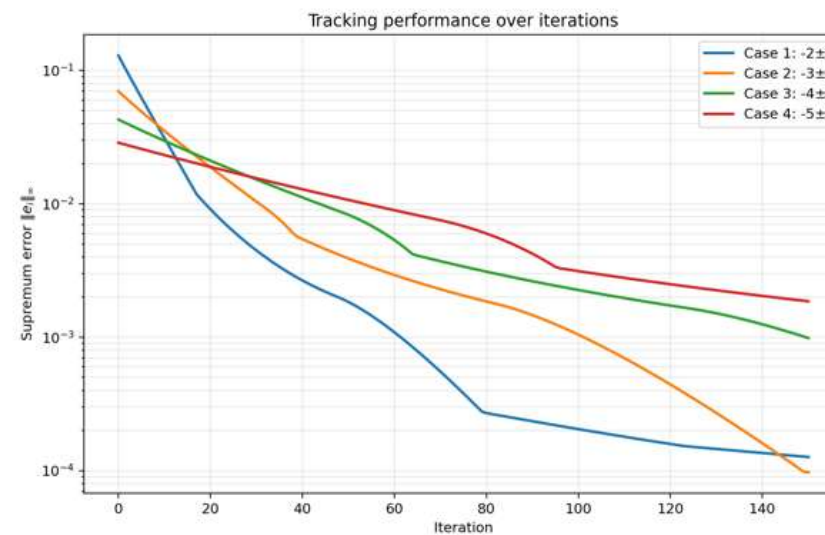
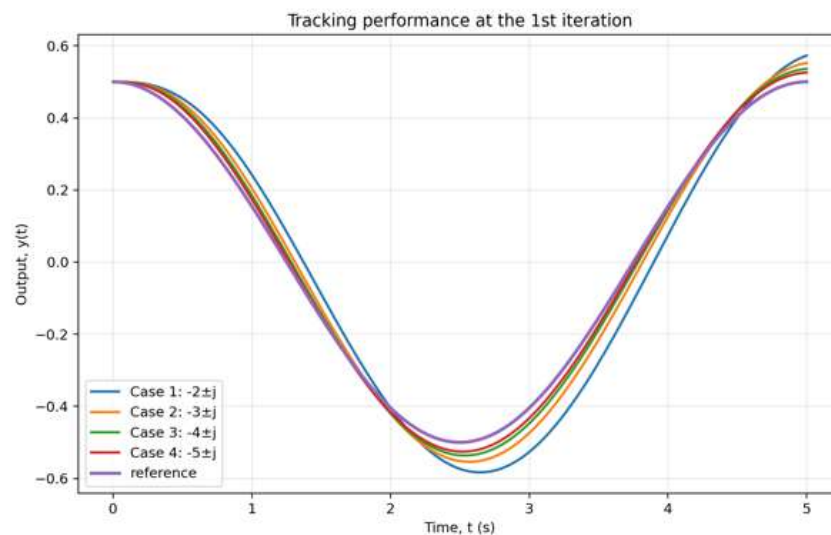
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Learnability and PE Condition using Input Learnability

- The convergence proof does not explicitly reveal the effect of the feedback gain K on the convergence rate
- Let us select four different gains

	Case-1	Case-2	Case-3	Case-4
CL- poles	$-2 \pm j$	$-3 \pm j$	$-4 \pm j$	$-5 \pm j$
K	$[6 \ 4]$	$[11 \ 6]$	$[18 \ 8]$	$[27 \ 10]$

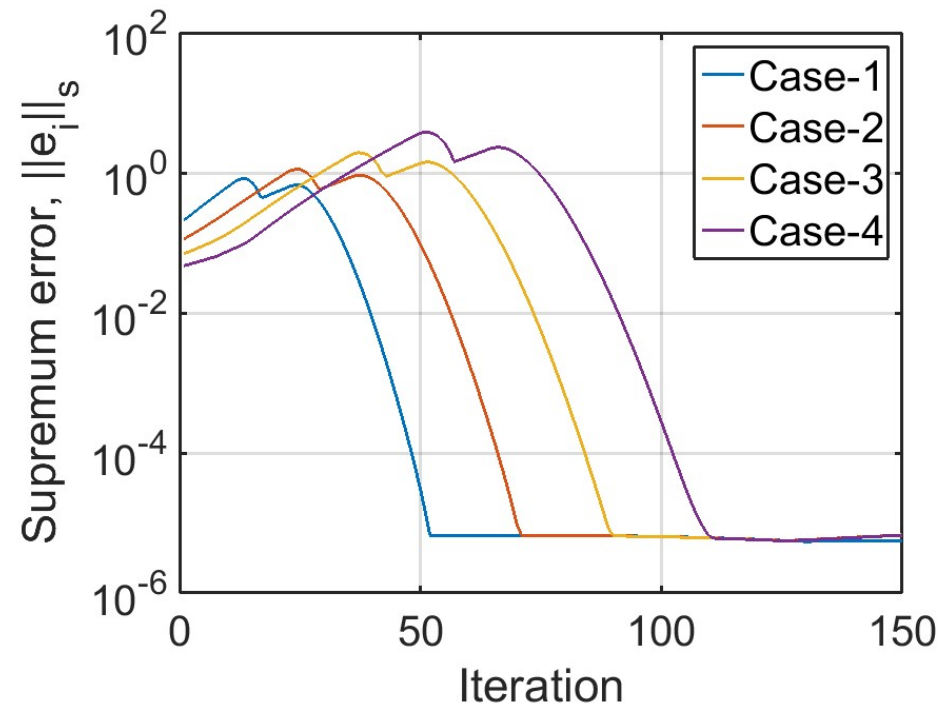
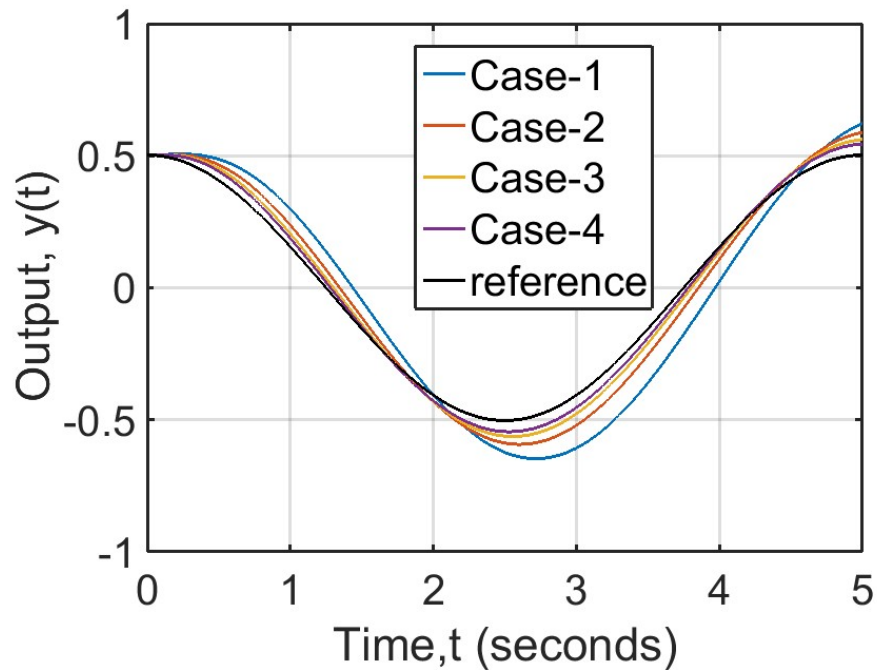




Learnability and PE Condition using Input Learnability

➤ Our simulation results showed that: YT1

$$\|e_i\|_s = \max_{t \in [0, T]} |e_i(t)|$$



Slide 19

YT1

Jonathan, can you please generate this figure for me?

Ying Tan, 2026-08-20T01:13:43.278



Learnability and PE Condition using Input Learnability

➤ Let us explain the simulation results from PE condition

$$\Delta \dot{x}_i(t) = (A - BK)\Delta x_i(t) + B(u_d(t) - u_i^H(t))$$

$$\Delta x_i(t) = \int_0^t \Phi(t - \tau) B(u_d(\tau) - u_i^H(\tau)) d\tau \quad \Phi(t) = e^{(A-BK)t}$$

$$\dot{e}_{i+1}(t) = -B^T \int_0^t \Phi(t - \tau) B (\Delta u_{i+1}^H(\tau)) d\tau = -K \Delta u_{i+1}^H(t)$$

$$u_{i+1}^H(t) = u_i^H(t) + \dot{e}_{i+1}(t)$$

$$\Delta u_{i+1}^H(t) = \Delta u_i^H(t) + \dot{e}_{i+1}(t) = \Delta u_i^H(t) - K \Delta u_{i+1}^H(t)$$



Learnability and PE Condition using Input Learnability

- Let us define the energy related to input learning

$$E_i^H = \|\Delta u_i^H\|_{L^2[0,T]}^2 = \int_0^T (\Delta u_i^H(t))^2 dt$$

$$(1 + K)\Delta u_{i+1}^H \Delta u_{i+1}^H(t) = \Delta u_i^H(t)$$

$$E_{i+1}^H + 2\langle K\Delta u_{i+1}^H(t), \Delta u_{i+1}^H(t) \rangle + \|K\Delta u_i^H\|_{L^2[0,T]}^2 = E_i^H$$

- Here K denotes an operator

$$K\Delta u_{i+1}^H(t) = \int_0^t B^T \Phi(t - \tau) B (\Delta u_{i+1}^H(\tau)) d\tau$$



Learnability and PE Condition using Input Learnability

➤ Let us define a PE-inspired condition.

$$\langle \Delta u_{i+1}^H, K \Delta u_{i+1}^H \rangle \geq \alpha \|\Delta u_{i+1}^H\|_{L^2[0,T]}^2$$

$$E_{i+1}^H + 2 \langle K \Delta u_{i+1}^H(t), \Delta u_{i+1}^H(t) \rangle + \|K \Delta u_i^H\|_{L^2[0,T]}^2 = E_i^H$$

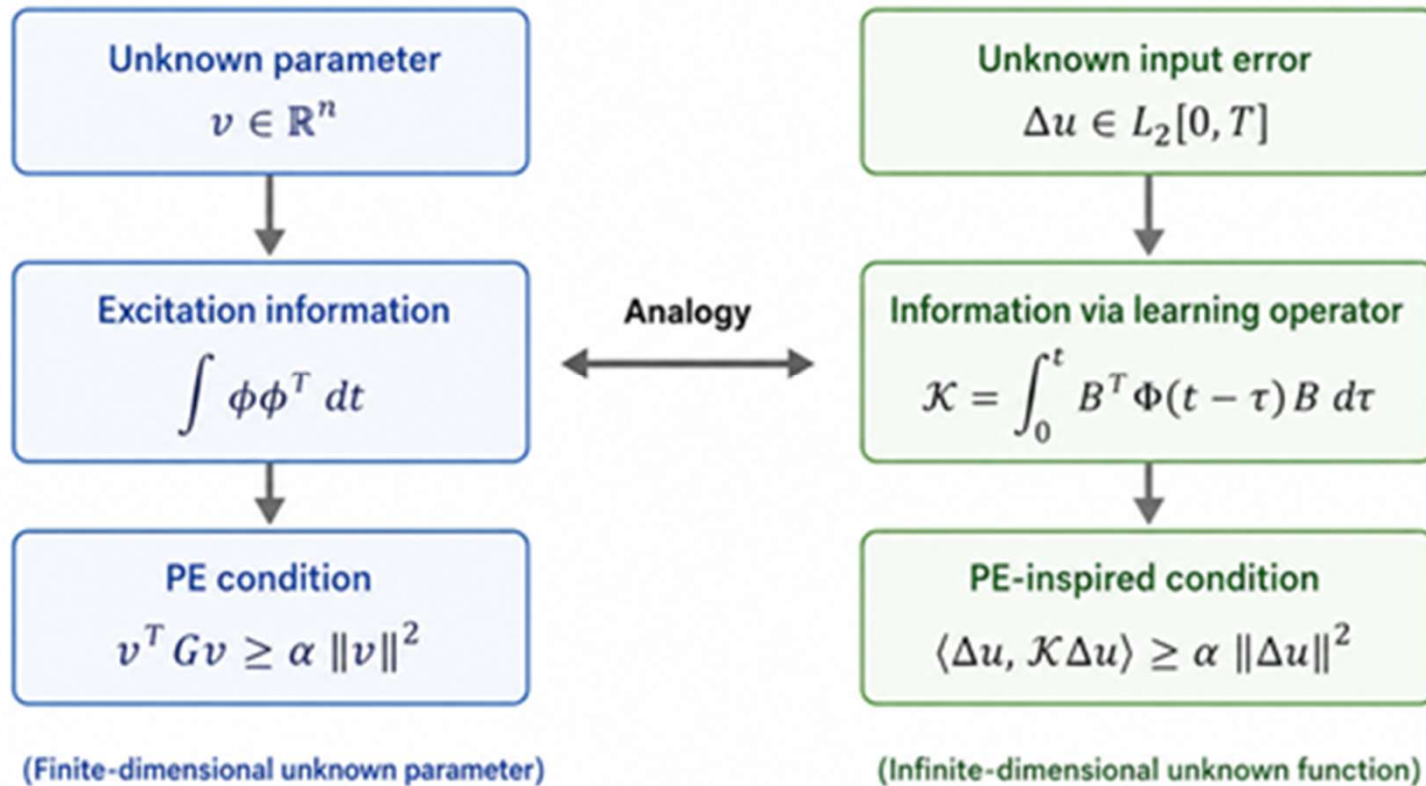
$$E_{i+1}^H (1 + 2\alpha) + \|K \Delta u_i^H\|_{L^2[0,T]}^2 \leq E_i^H$$

$$E_{i+1}^H (1 + 2\alpha) \leq E_i^H$$

$$E_{i+1}^H \leq \frac{1}{1+2\alpha} E_i^H$$

Learnability and PE Condition using Input Learnability

- A larger α implies a faster guaranteed convergence rate





Summary

1. We introduce a PE-inspired condition for input learnability
2. PE characterizes not only learnability but also the convergence rate
3. Feedback does more than stabilize the system—it shapes the excitation available for learning



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Thank you